

The background features a blue gradient with a series of overlapping, semi-transparent geometric shapes (triangles and squares) in various shades of blue and grey. On the right side, there is a vibrant, multi-colored particle burst or explosion effect, with colors ranging from purple and pink to yellow and green. The text is centered in the lower half of the image.

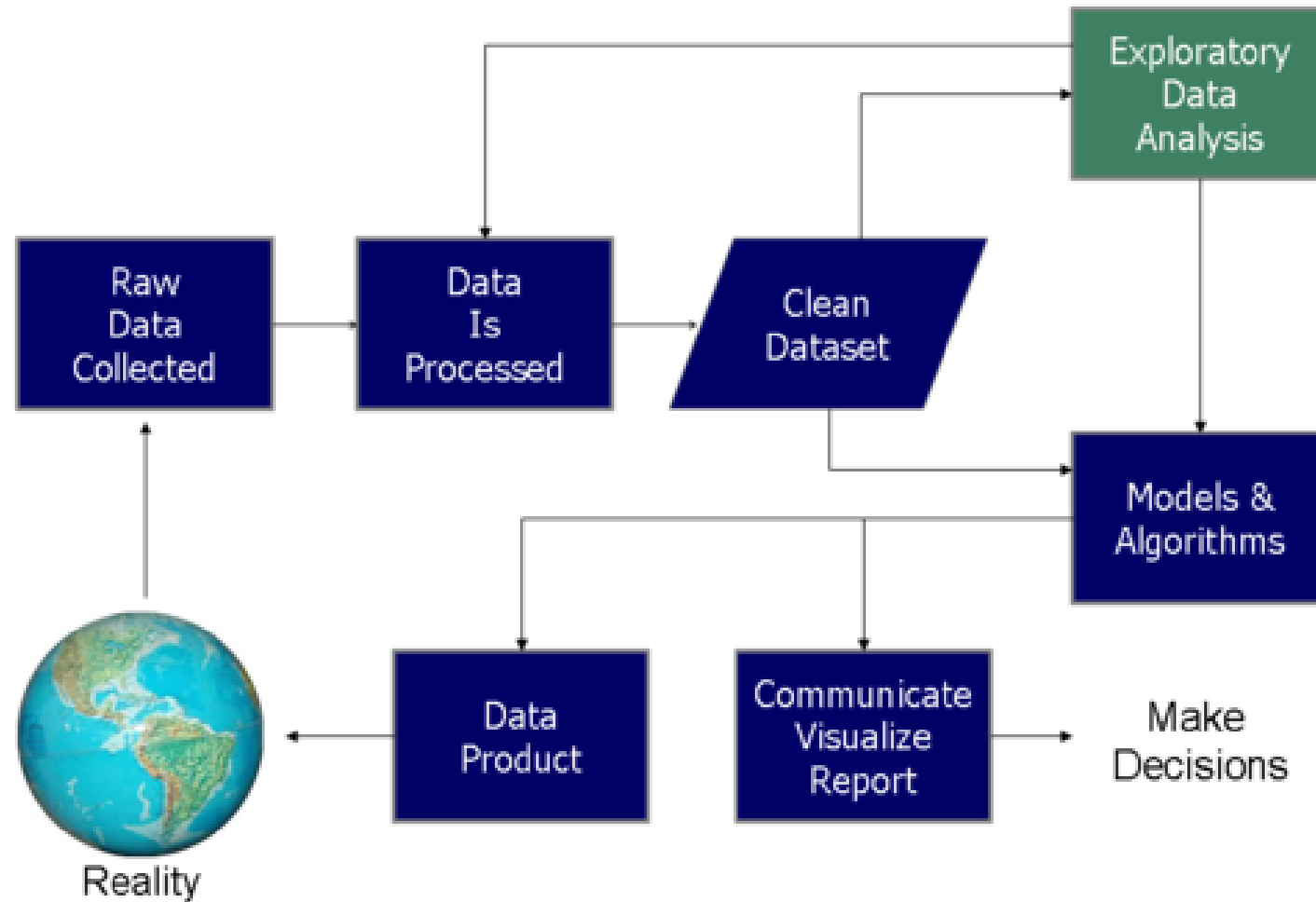
# TEHNICI DE COLECTARE SI MANIPULARE A DATELOR

# Evaluarea studentilor

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| Activitatea          | Procent |
|----------------------|---------|
| Examen               | 70%     |
| Activitate laborator | 20%     |
| Prezenta             | +10%    |
| Proiect              | 10%     |
| Curs AWS Academy     | +20%    |

# Data Science Process



# THE DATA SCIENCE PROCESS



Data Engineers

Data Analysts

Machine Learning Engineers

Data Scientists



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ChatGPT is a large language conversational AI built by OpenAI. It was trained using Reinforcement Learning from Human Feedback, similar to InstructGPT. ChatGPT understands the prompt and provides detailed response that can help you with research, coding, and various data science tasks.

## Ideas

### Dataset Suggestion

\*\*\* I want to build a predictive model for image classifiers. Can you please suggest the five most relevant datasets for my use case?

### Suggest Resources

\*\*\* I would like to learn about deep learning. Please suggest 3 best specific resources.

### AB Testing

\*\*\* [case-study]. Please design an A/B test for this purpose. Please include the concrete steps on which statistical test I should run.

### Career Coaching

\*\*\* I am looking for a role as a data engineer. My background is management. What should I do in 6 months to get a job?

## Coding

### Unit Test

\*\*\* Write a unit test for train function. The test cases are: x should not be null value and y should be a numerical value.

### Code Explanation

\*\*\* Can you explain what the code is doing? [code-snippet]

### Optimize Code

\*\*\* Can you improve the time complexity of the code? [code-snippet]

## SQL

### SQL Formatting

\*\*\* Format the following SQL code and convert all reserved keywords to uppercase. [code-snippet]

### Translate Between DBMS

\*\*\* What is the equivalent of PostgreSQL's DATE\_TRUNC for MySQL?

### Calculate Average

\*\*\* Write the SQL code that works for PostgreSQL 14. I have a table with two columns [date, temp]. I would like to calculate an average temp.

### Calculate Runway

\*\*\* Write SQL to calculate my runway.

## Spreadsheets

### Spreadsheets Formula

\*\*\* Create a spreadsheet formula to calculate the sum of cells B1 to B20?

### Dummy Data

\*\*\* Generate the dummy data for me to use as placeholders in my spreadsheet.

### Tips

\*\*\* Give me some tips on how to improve the efficiency of my spreadsheet?

## Data Analysis

### Generate Data

\*\*\* Generate a fake data with 100 rows and 4 columns: [id,name,grade,subject]

### Data Cleaning

\*\*\* I have a text classification dataset. Write Python code for data cleaning.

### Data Exploration

\*\*\* I have a dataset of 100 rows and four columns[id, name, grade, subject]. Write R code for data visualization and exploration.

### Data Visualization

\*\*\* I have a dataset with 100 rows columns [id, name, grade, subject]. Create a matplotlib bar chart of subject vs. grade.

## Machine Learning

### Train Regression Model

\*\*\* You are a data scientist, write Python code for me. I have a dataset with columns [model,hp,speed]. Please build a machine learning model that predicts speed.

### Hyperparameter Tuning

\*\*\* I have a logistic regression model, write Python code to tune hyperparameters.

### Imbalance Data

\*\*\* I have an imbalanced dataset with target column species. In python, how do I oversample and/or undersample my data?

### Explain the Model

\*\*\* I have trained a LightGBM model. Write a Python code to explain the output using a series of plots with Shap.

## Research

### Explain the Concept

\*\*\* Explain t-test to an undergraduate as a data science instructor.

### Stakeholders

\*\*\* Tips on explaining data science reports to a business stakeholder.

### Summarize the paper

\*\*\* Please summarize the paper "Adding Conditional Control to Text-to-Image Diffusion Models" in simple terms in one paragraph.

### Writing Blog

\*\*\* Write an outline for a blog "Python lists".

### Research History

\*\*\* Can you research the history of the graph neural network?

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# Continut

## Introducere

- Pregatirea datelor
- Librarii Python dedicate stiintei datelor
  - Numpy
  - Pandas
  - Matplotlib
  - ScyPy
  - sklearn (scikit-learn)

## Colectarea datelor

## Curatarea/manipularea datelor

- Pregatirea datelor
- Tratarea datelor lipsa
- Transformarea datelor
- Aplicarea tehnicilor de normalizare a datelor

## Explorarea datelor

- Vizualizarea datelor
- Analiza statistica a datelor

# Laborator

- Anaconda
- una din cele mai populare platforme pentru data science
- free, open-source environment
- <https://www.anaconda.com/>
- IPython and Jupyter Notebook
- Contine peste 7500 colectii de pachete/librarii scrise in Python/R



# Colectii de date

- kaggle
- <https://www.kaggle.com/datasets>
- UCI Machine Learning Repository
- <https://archive.ics.uci.edu/ml/datasets.php>
- AWS
- [Open Data on AWS](#)
- [data.gov.ro](http://data.gov.ro)
- [Institutul Național de Statistică \(insse.ro\)](http://insse.ro)
- <http://www.ecad.eu> – European Climate Assessment & Dataset

- <https://nomenclator.anm.ro/medicamente>
- <https://www.re3data.org/>
- <https://databrowser.uis.unesco.org/>
- Research Data Alliance in Romania
- Data sweep

# Bibliografie recomandata

- Wes McKinney, *Python for Data Analysis*, O'Reilly 2018
- Jake VanderPlas, *Python Data Science Handbook*, O'Reilly 2017
- Maria C. Mariani, Maria Pia Beccar-Varela, Osei Kofi Tweneboah, *Data Science in Theory and Practice*, John Wiley & Sons, 2022
- Etc.

# Webografie recomandata

- <https://www.kaggle.com>
- <https://aws.amazon.com/training/awsacademy/>
- Etc.